

Classification Metrics

Homework #2

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1 Homework Solutions

1. Download the classification output data set (attached in Blackboard to the assignment).

```
classifications_output_data <- read.csv("classification-output-data (2).csv")  
  
classifications_output_data |> skim()
```

Table 1: Data summary

Name	classifications_output_da...
Number of rows	181
Number of columns	11
Column type frequency:	
numeric	11
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
pregnant	0	1	3.86	3.24	0.00	1.00	3.00	6.00	15.00	
glucose	0	1	118.30	30.48	57.00	99.00	112.00	136.00	197.00	
diastolic	0	1	71.70	11.80	38.00	64.00	70.00	78.00	104.00	
skinfold	0	1	19.80	15.69	0.00	0.00	22.00	32.00	54.00	
insulin	0	1	63.77	88.73	0.00	0.00	0.00	105.00	543.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
bmi	0	1	31.58	6.66	19.40	26.30	31.60	36.00	50.00	
pedigree	0	1	0.45	0.28	0.09	0.26	0.39	0.58	2.29	
age	0	1	33.31	11.18	21.00	24.00	30.00	41.00	67.00	
class	0	1	0.31	0.47	0.00	0.00	0.00	1.00	1.00	
scored.class	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
scored.proba- bility	0	1	0.30	0.23	0.02	0.12	0.24	0.43	0.95	

2. The data set has three key columns we will use:

- *class*: the actual class for the observation
- *scored.class*: the predicted class for the observation (based on a threshold of 0.5)
- *scored.probability*: the predicted probability of success for the observation

Use the `table()` function to get the raw confusion matrix for this scored dataset. Make sure you understand the output. In particular, do the rows represent the actual or predicted class? The columns?

```
confusion_matrix <- table(classifications_output_data$class, classifications_output_data$scored.class)
confusion_matrix
```

```
      0   1
0 119   5
1  30  27
```

The rows represent the actual class (`class`) and the columns represent the predicted class (`scored.class`). For example, there are 119 observations where the actual class is 0 and the predicted class is also 0 (true negatives), and there are 30 observations where the actual class is 1 but the predicted class is 0 (false negatives).

3. Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the accuracy of the predictions.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

```
calculate_accuracy <- function(data, actual_col, predicted_col) {
  confusion_matrix <- table(data[[actual_col]], data[[predicted_col]])
  tp <- confusion_matrix[2, 2]
```

```

    tn <- confusion_matrix[1, 1]
    fp <- confusion_matrix[1, 2]
    fn <- confusion_matrix[2, 1]
    accuracy <- (tp + tn) / (tp + fp + tn + fn)
    return(accuracy)
}

# Test function
calculate_accuracy(classifications_output_data, "class", "scored.class")

```

```
[1] 0.8066298
```

4. Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the classification error rate of the predictions.

$$\text{ClassificationErrorRate} = \frac{FP + FN}{TP + FP + TN + FN}$$

Verify that you get an accuracy and an error rate that sums to one.

5. Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the precision of the predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

6. Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the sensitivity of the predictions. Sensitivity is also known as recall.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

7. Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the specificity of the predictions.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

8. Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the F1 score of the predictions.

$$\text{F1Score} = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}$$

9. *Before we move on, let's consider a question that was asked: What are the bounds on the F1 score? Show that the F1 score will always be between 0 and 1. (Hint: If $0 < p < 1$ and $0 < r < 1$ then $pr < .$)*
10. *Write a function that generates an ROC curve from a data set with a true classification column (class in our example) and a probability column (scored.probability in our example). Your function should return a list that includes the plot of the ROC curve and a vector that contains the calculated area under the curve (AUC). Note that I recommend using a sequence of thresholds ranging from 0 to 1 at 0.01 intervals.*
11. *Use your created R functions and the provided classification output data set to produce all of the classification metrics discussed above.*
12. *Investigate the caret package. In particular, consider the functions confusionMatrix, sensitivity, and specificity. Apply the functions to the data set. How do the results compare with your own functions?*
13. *Investigate the pROC package. Use it to generate an ROC curve for the data set. How do the results compare with your own functions?*

2 Classification Metrics

Upon following the instructions below, use your created R functions and the other packages to generate the classification metrics for the provided data set.